

Kitchcu

Complete Executive Guide - CEO, CPO & CTO

Kitchcu cloud kitchen platform

Audience: CEO, CPO, CTO, Product, Engineering, Investors

Three executive lenses:

- CEO - vision, market, business model, GTM, metrics, risks
- CPO - personas, pains, 48 features, modules, flows, KPIs
- CTO - architecture, services, DB, events, APIs, build status

This guide includes:

- Full strategic positioning vs food aggregators and POS systems
- 48-feature catalog with implementation status (Sprints S1-S4 backend complete)
- Owner and customer pain points mapped to modules and live code
- Event-driven microservices, outbox pattern, and API reference
- Subscription tiers, unit economics, and development phases
- 90+ automated tests; PWAs on customer.kitchcu.in / kitchen.kitchcu.in / admin.kitchcu.in
- Marketing portal with parallax UI, AI support chat, and admin ticketing

Table of Contents

PART I - CEO Lens

1. Executive Summary
2. Market & Strategic Positioning
3. Business Model & Unit Economics
4. Go-to-Market & Development Phases
5. North-Star Metrics & Investment Thesis
6. Risks & Mitigations

PART II - CPO Lens

7. Product Vision & Personas
8. Pain Points to Solutions
9. Platform Modules & Feature Catalog
10. Application Flows
11. Product Principles & KPIs

PART III - CTO Lens

12. Architecture Overview
13. Services, Events & Data Flow
14. Database & Caching
15. API Reference & Security
16. Build Status & Engineering Standards

Appendices

- A. Feature Implementation Matrix
- B. Document Index

Executive Strategy

Kitchcu is a B2B2C cloud kitchen operating system - not a food aggregator. It gives kitchen owners full control over orders, quality, marketing, and customer relationships while giving diners real-time transparency into their food.

Stakeholder	Pain Today	Kitchcu Answer
Owner	WhatsApp chaos, commissions	Unified hub, zero commission
Customer	Stock photos, opaque delivery	Live-capture, fair fees
Market	No quality benchmark	Chef rankings (Phase 3)

"Keep the product simple for the owner on day one. First WhatsApp order in under 5 minutes of onboarding."

2. Market & Strategic Positioning

Aggregators	vs	Kitchcu
Per-order commission		Monthly subscription
Platform owns customer		Owner owns CRM
Stock photos		Live-capture media
Speed-first delivery		Quality-first SLA
Single kitchen per cart		Multi-kitchen cart

Competitive moat: WhatsApp-native ops + live-capture integrity + owner-owned CRM + quality-first lifecycle.

3. Business Model & Unit Economics

Tier	Monthly	Includes
Starter	Rs 499	1 kitchen, WhatsApp, basic reports
Pro	Rs 1,499	CRM, coupons, tiffin, marketing
Enterprise	Rs 3,999	Live stream, API, branches

Zero per-order food commission. Kitchen keeps 100% of food revenue.

Metric	Year 1 Goal
CAC (owner)	< Rs 2,000
Owner LTV	> Rs 18,000
LTV:CAC	> 3:1
Gross margin	> 75%

4. Go-to-Market & Development Phases

Phase	Timeline	Goal
1 Foundation	Mo 1-3	10 pilot kitchens, daily orders
2 Growth	Mo 4-6	Customer PWA, CRM, multi-kitchen
3 Differentiation	Mo 7-10	Rankings, live stream
4 Scale	Mo 11-12+	National platform, white-label

Phase 1 Milestones (Current)

Milestone	Deliverable	Status
M1.1	Identity, gateway, Docker	COMPLETE
M1.2	Catalog + live photo	COMPLETE
M1.3	Order + WhatsApp intake	COMPLETE
M1.4	Lifecycle + notifications	PARTIAL
M1.5	Owner/customer/admin PWAs + portal	S5 PARTIAL
M1.6	Razorpay billing	S6

5. North-Star Metrics & Investment Thesis

- Owner GMV managed on platform (primary north star)
- Repeat order rate (customer loyalty proxy)
- Dish rating consistency (quality standardization)
- Owner NPS (aggregator dependency reduction)

KPI	MVP	12-Month
Active kitchens	10	500
Orders/day	50	5,000
Owner retention	80%	90%
Repeat rate (30d)	25%	40%

Investment thesis: Kitchcu captures the operating system layer for India's cloud kitchens - subscription SaaS with zero food commission. Phase 1 proves order throughput; Phase 2 unlocks customer network effects; Phase 3 builds community moats. Capital-efficient PWA-first approach.

6. Risks & Mitigations

Risk	Mitigation
WhatsApp API changes	Manual input always available
Live stream complexity	Managed provider; opt-in only
Multi-kitchen disputes	Per-kitchen sub-order IDs
Scope creep	Strict MoSCoW; Phase 1 sacred
Onboarding friction	WhatsApp-first; 5-min setup

Product Depth

Empower every cloud kitchen to scale like a brand - with data, quality standards, and direct customer relationships - while giving diners honest visibility into their food.

Persona	Goals	Surface
Owner/Chef	Orders, quality, marketing	kitchen.kitchcu.in PWA
Customer	Discover, order, rate	Customer PWA (Ph 2)
Staff	Lifecycle, inventory	Staff PWA subset
Admin	Moderate, support tickets	admin.kitchcu.in

PWA-First Strategy

- One codebase: Android, iOS, desktop
- WhatsApp links open directly - critical for order capture
- Offline: menu, orders, draft queue via service workers
- Live dish photo via getUserMedia in browser
- Target: Lighthouse PWA score >= 90

2. Pain Points to Solutions

Owner Pains (P1-P8)

#	Pain	Module	Status
P1	WhatsApp chaos	Order+Notify	Partial
P2	Aggregator commission	Billing	S6
P3	No profit visibility	Analytics	Partial
P4	Stock photo deception	Catalog	Done
P5	Taste inconsistency	Catalog	Partial
P6	No owner CRM	Marketing	Ph 2
P7	Promotion guesswork	Growth	Ph 2
P8	Multi-channel chaos	Order	Partial

Customer Pains (C1-C6)

#	Pain	Solution	Status
C1	Untrustworthy photos	Live-capture media	Done
C2	Opaque delivery fees	PostGIS + rules	Fields only
C3	No tracking	Lifecycle + notify	Partial
C4	Generic ratings	Home-taste score	Ph 2
C5	Single-kitchen cart	Multi-kitchen cart	Ph 2
C6	No tiffin	Meal plans	Ph 2

3. Platform Modules & Feature Catalog

Module	Schema	Sprint	Status
Gateway	-	S1	Live
Identity	ckac_identity	S1	Live
Catalog	ckac_catalog	S2	Live
Order	ckac_orders	S3	Live
Notification	ckac_support	S4	Live
Billing	ckac_billing	S6	Planned
Analytics	order svc	S5	Partial

Marketing	ckac_marketing	Ph 2	Planned
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48 Features by Phase

- Phase 1 (Must): F01-F05, F07, F13-F15, F26, F30, F42-F43, F45
- Phase 2 (Growth): F06, F08-F12, F16-F18, F25, F27-F33, F34-F41, F44
- Phase 3 (Differentiation): F19-F24, F46-F48
- Done: 8 features | Partial: 4 | Total spec: 48

4. Application Flows

Owner Day-1 Onboarding

Register -> OTP -> JWT -> Create kitchen (CKPNQ001)
-> Add dishes (live photo required)
-> Connect WhatsApp -> First order in inbox

Order Lifecycle (F04)

received -> accepted -> preparing -> ready
-> out_for_delivery -> delivered
(cancelled with reason at any stage)

WhatsApp Order (F01)

Meta webhook -> notification (lookup kitchen)
-> whatsapp.message.received event
-> order internal API -> parse -> draft
-> owner confirms -> order.placed

Multi-Kitchen Checkout (F06, Phase 2)

Cart: Kitchen A + Kitchen B items
-> master_orders (1 payment)
-> order A + order B (separate lifecycle)
-> Razorpay Route split settlement

5. Product Principles & KPIs

- WhatsApp-native order intake
- Quality over speed - owner-set prep windows
- Trust through media - live capture required
- Owner-owned CRM - customer data belongs to kitchen
- Progressive complexity - MVP first, growth layers later

Metric	Phase 1 Target
Time to first order	< 5 min
Owner daily active	80% paying kitchens
Order capture rate	95% parsed or manual
Menu trust score	100% live-capture heroes
Owner retention M6	80%

Technical Architecture

Experience: kitchcu.in portal | customer.kitchcu.in | kitchen.kitchcu.in | admin.kitchcu.in

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Edge: API Gateway :18000 (JWT, routing)

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Application: identity | catalog | order | notification

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Data: PostgreSQL 16 + PostGIS | Redis Streams + cache

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Media: MinIO / S3 (live-capture URLs)

Event-driven microservices on Python 3.12 + FastAPI. Schema-per-bounded-context. Transactional outbox on all writes.

Layer	Technology
API	Python 3.12, FastAPI, Pydantic v2
Database	PostgreSQL 16 + PostGIS
Events/Cache	Redis 7 Streams + menu cache 300s
Frontend	React + Vite + TypeScript PWA
Payments	Razorpay (S6)
WhatsApp	Meta Business Cloud API
Infra	Docker Compose -> Kubernetes

2. Services, Events & Data Flow

Service	Port	Key Events
gateway	18000	-
identity	18001	kitchen.created
catalog	18002	dish.created/updated
order	18003	order.placed, status.changed
notification	18005	whatsapp.message.received, support.ticke

Event Catalog

Event	Producer	Stream
kitchen.created	identity	ckac:identity:kitchen
dish.created/updated	catalog	ckac:catalog:dish
order.placed	order	ckac:orders:order
order.status.changed	order	ckac:orders:order
order.draft.created	order	ckac:orders:draft
whatsapp.message.received	notification	ckac:notify:whatsapp

Outbox Pattern (Critical)

- Domain write + outbox in same DB transaction
- flush_pending() after commit - no Redis-before-commit bug
- Relay worker for retry / Kafka migration - Phase 4
- pg_advisory_xact_lock for kitchen codes and bill IDs

3. Database & Caching

Schema	Tables	Status
ckac_identity	owners, kitchens	LIVE
ckac_catalog	categories, dishes, dish_media	LIVE
ckac_orders	orders, items, events, drafts	LIVE
ckac_events	outbox, processed_events	LIVE
ckac_support	support_tickets, messages	LIVE
ckac_billing	payments, subscriptions	S6
ckac_marketing	customers, coupons	Ph 2

Naming Conventions

Kitchen code: CKPNQ001

Bill ID: BILL-20260712-0001

Order code: CKPNQ001-BILL-20260712-0001

Cache Keys

Key	TTL	Invalidation
menu:{kitchen_id}	5 min	dish.updated
dish:{dish_id}	10 min	rating update
kitchen:{id}:profile	15 min	settings change

SLO Targets

Metric	Target
API p95 read	< 200 ms
API p95 write	< 500 ms
Order E2E	< 3 s
WhatsApp notify	< 10 s from event
Uptime	99.9% (Ph 2+)

4. API Reference & Security

Gateway Routes (:18000/api/v1)

Identity: /auth/* /owners/* /kitchens

Catalog: /kitchens/{id}/categories|menu|dishes

Order: /kitchens/{id}/orders/* /orders/{id}/status /analytics/*

Notification: /webhooks/whatsapp /support/chat /support/tickets

Admin: /admin/tickets/* (notification service)

Internal: /internal/... (X-Internal-Key)

Security

Area	Implementation
Auth	JWT + OTP; Bearer on owner routes
Tenant isolation	kitchen_id + RLS (prod)
Media	Live-capture validator; signed URLs
Internal	X-Internal-Key header
WhatsApp	Verify token required outside dev
Health	/health/live + /health/ready

5. Build Status & Engineering Standards

Tests passing	Backend + PWAs p	Services + gateway	Features spec
90+	S1-S5	5+1	48

Sprint	Deliverable	Status
S1	Identity, gateway, JWT	COMPLETE
S2	Catalog, live-capture, EDD	COMPLETE
S3	Order, lifecycle, history	COMPLETE
S4	WhatsApp webhook + parser	COMPLETE
S5	PWAs + analytics + support/tickets	PARTIAL
S6	Billing + revenue	Planned

Engineering Standards

- TDD: RED -> GREEN -> REFACTOR for every feature
- EDD: DB commit first; session-bound EventPublisher
- Service template: main, routes, schemas, models
- No cross-schema writes; Alembic per service
- Run tests: scripts/run-tests.ps1

Reference Tables

ID	Feature	Status
F01	WhatsApp capture	Partial
F02	Message parser	Done
F03	Manual order	Done
F04	Lifecycle	Done
F05	Order history	Done
F07	Revenue report	Partial
F13	Dish + live photo	Done
F14	Price/quality fields	Done
F15	Categories	Done
F26	Subscription	Partial
F30	Prep time	Done
F42-F43	Payments	S6
F45	Notify + AI chat + tickets	Partial

2. Document Index

Document	Audience
CKAC-COMPLETE-GUIDE.md	CEO, CPO, CTO (this guide)
CKAC-IMPLEMENTATION-GUIDE.md	Engineering (live code map)
CKAC-ARCHITECTURE-CTO.md	CTO, EM (layers + traceability)
CKAC-SYSTEM-BENCHMARK.md	CTO, DBA (deep spec)
CKAC-CPO-PRODUCT-BLUEPRINT.md	CPO, Product
CKAC-PITCH-DECK.pdf	CEO, Investors (33 slides)
AGENTS.md	Developers, AI agents

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